

IOWA STATE UNIVERSITY



AERE 4210: ADVANCED FLIGHT STRUCTURES

FALL 2024

DR. BRANDON RUNNELS

Lab Assignment 7

SECTION A GROUP 2

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1 Introduction

Vibration analysis for structures is an important analysis technique. While vibrations in structures are normal and expected, they can cause structural failure. By analyzing the resonance and natural frequencies of a structure, we can understand more about how it will behave and if it may fail under certain vibrations.

Part one involves a vibrational analysis of a tuning fork. Part two of the lab analyzes a hollow boom-skin structure through vibrational and harmonic analysis.

2 Data and Analysis

The tuning fork is first analyzed with geometry as describe in the problem statement, and later the length of the tuning fork is redesigned to achieve a certain frequency. Resonant modes of the fork are found as well as the ideal tuning fork length to have a first resonant frequency at concert A.

2.1 Part 1

Vibration Analysis

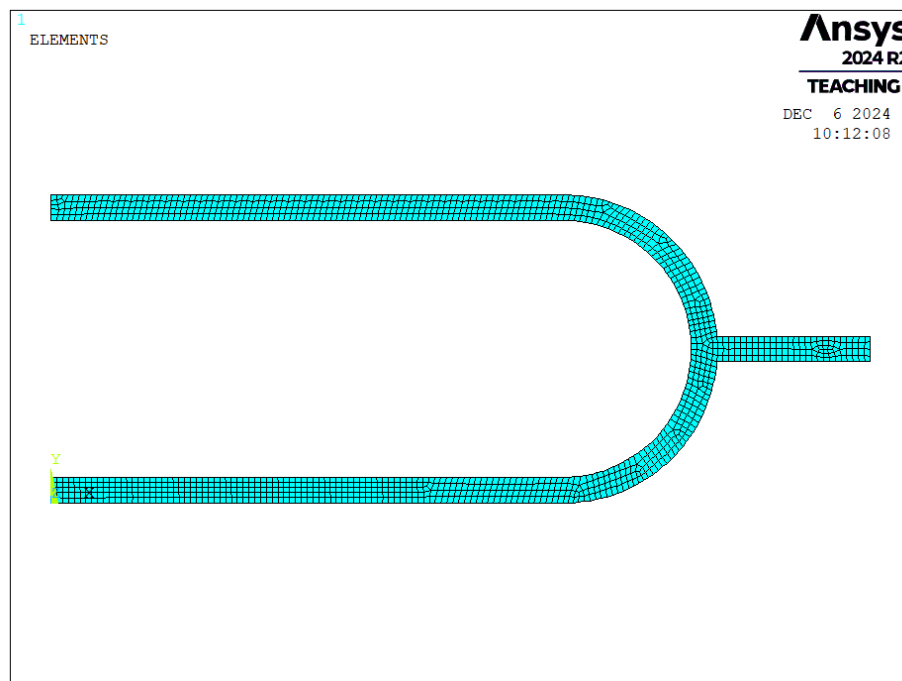


Figure 1: Part 1 Geometry.

Vibration analysis was performed on the tuning fork (Figure 1). The mode frequencies are tabulated below.

Table 1: Part 1 Natural Frequencies.

Mode	Frequency [Hz]
1	216.59759
2	689.85359
3	1392.9900
4	2216.5049
5	3614.5170
6	3995.4678
7	5347.4005
8	6567.3612
9	9625.7006
10	10800.454

Visualizations of various mode shapes are also shown below. Note that not all modes are shown, but the modes shown are varying modes with distinct shapes.

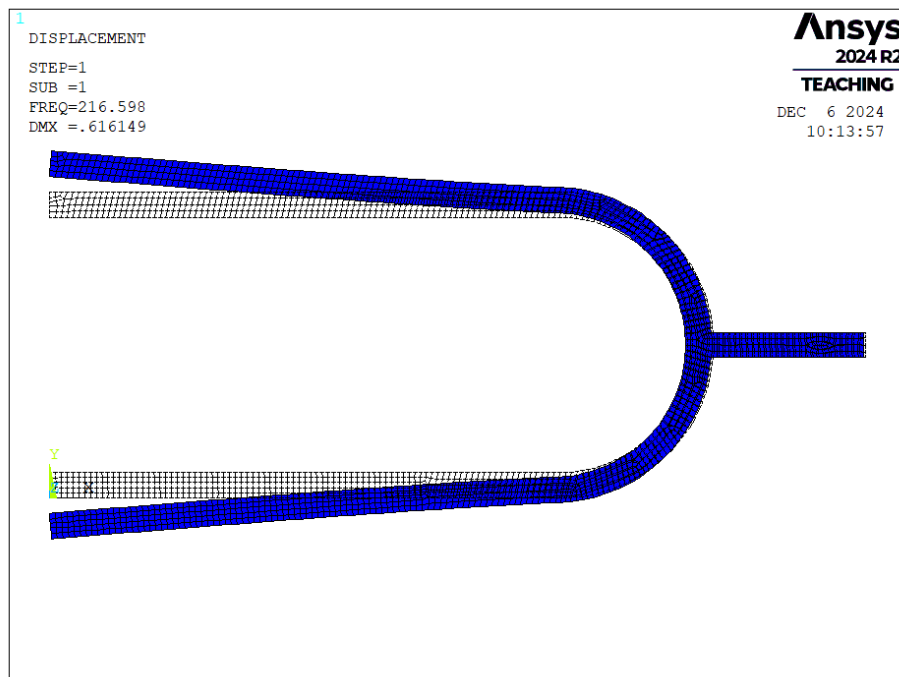


Figure 2: Mode 1 shape.

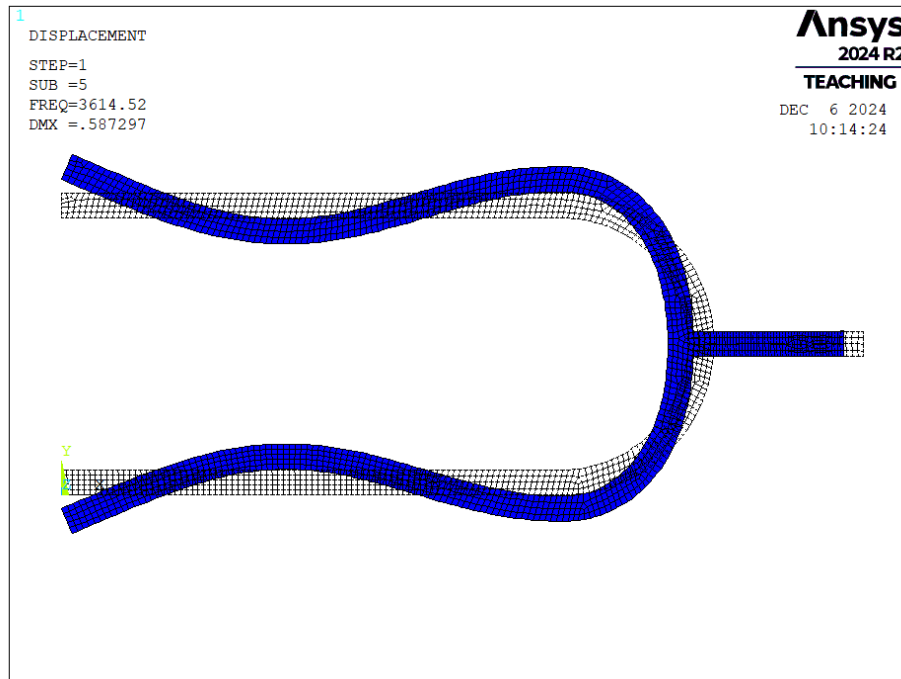


Figure 3: Mode 5 shape.

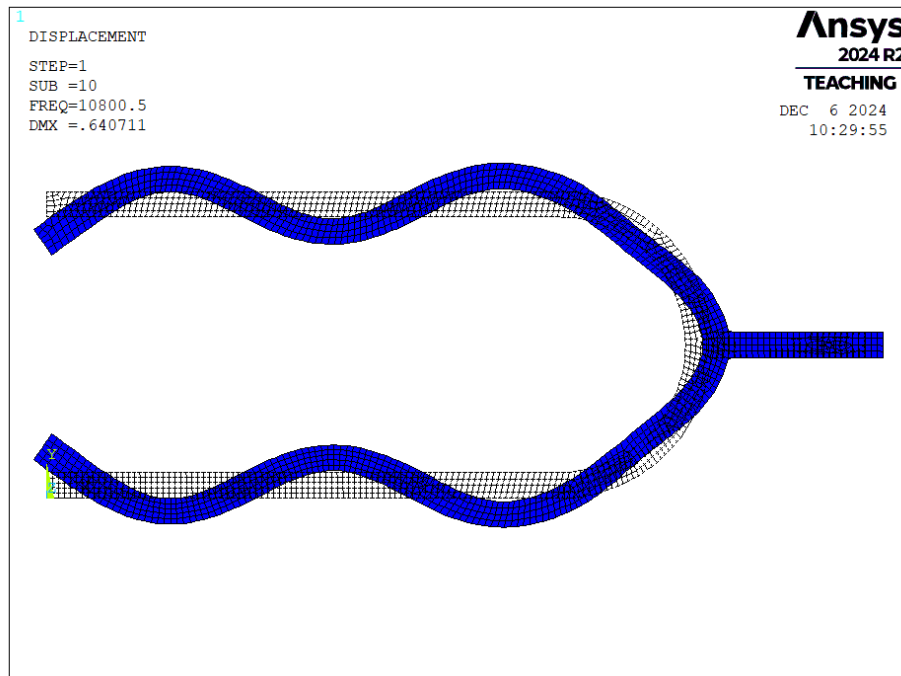


Figure 4: Mode 10 shape.

Design

After the initial vibration analysis, varying lengths for the tuning fork were testing to design a tuning fork with a first resonant frequency of 440 Hz (concert A). Note that this was not an optimization, rather just trying different values to get close to 440 Hz.

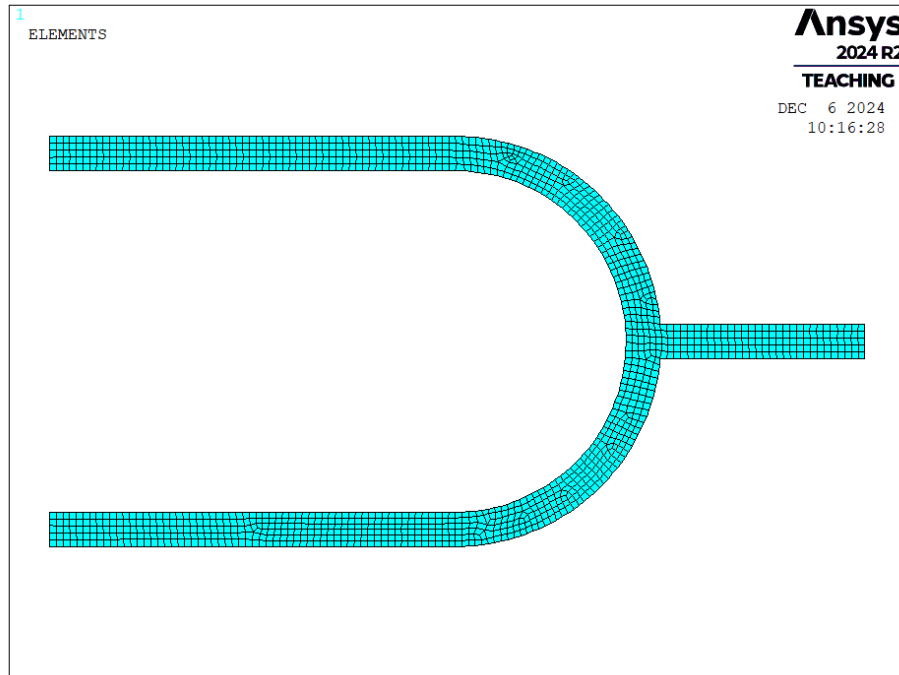


Figure 5: Part 1 440Hz design geometry.

In this analysis, the exact length of the straight section of the prongs of the tuning fork (from far left side to just before the fork starts curving) is **0.05945 m**. The exact frequency value, corresponding mode shape (Figure 6), and overtones of the fork that fall within human hearing range are shown below. The mode one frequency is 440.00975 Hz, which is well within one percent error of the desired value.

Table 2: Design Natural Frequencies.

Mode	Frequency [Hz]
1	440.00975
2	1244.1619
3	2728.0872
4	4055.3611
5	4771.7814
6	3995.4678
7	10888.911
8	12766.507
9	17898.935

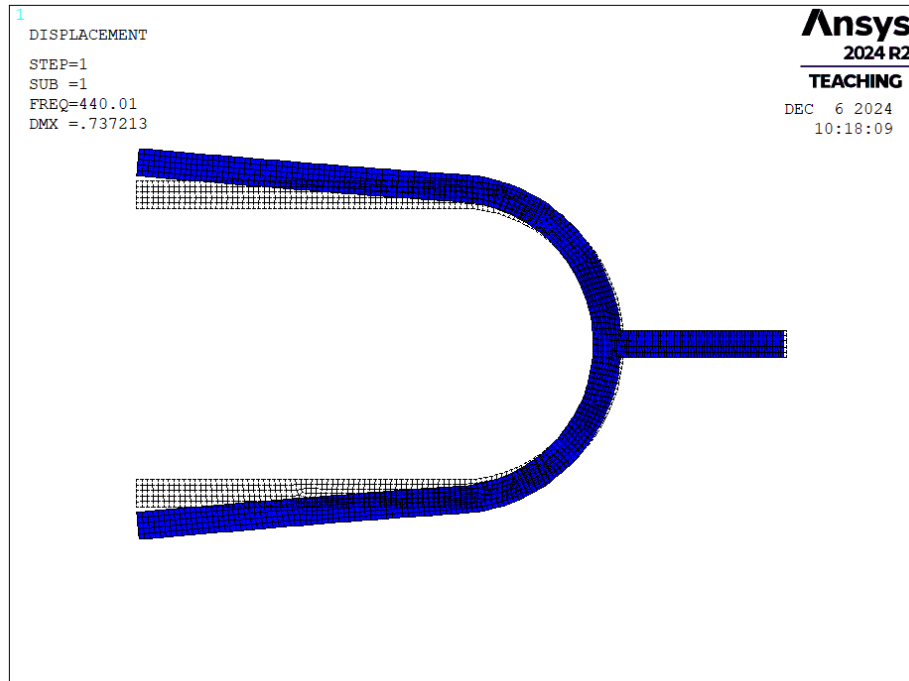


Figure 6: 440Hz design mode 1 shape.

2.2 Part 2

Problem 1: Vibration Analysis

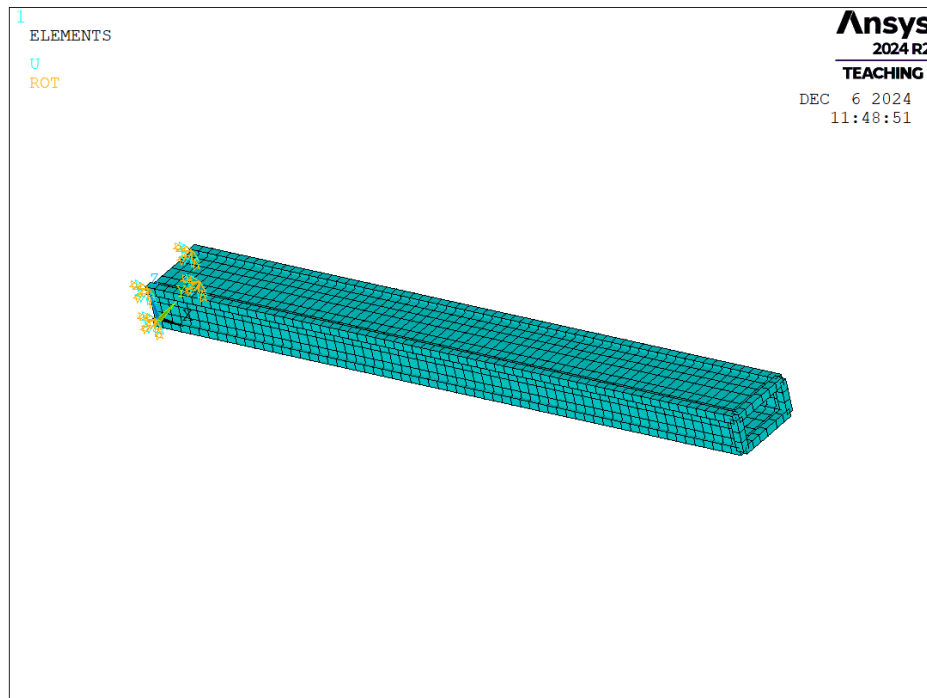


Figure 7: Part 2 geometry.

The hollow boom-skin structure (Figure 7) is first subjected to a vibration analysis. Visualization of the first five resonant mode shapes are shown below. Their respective natural frequencies are listed in Table 3.

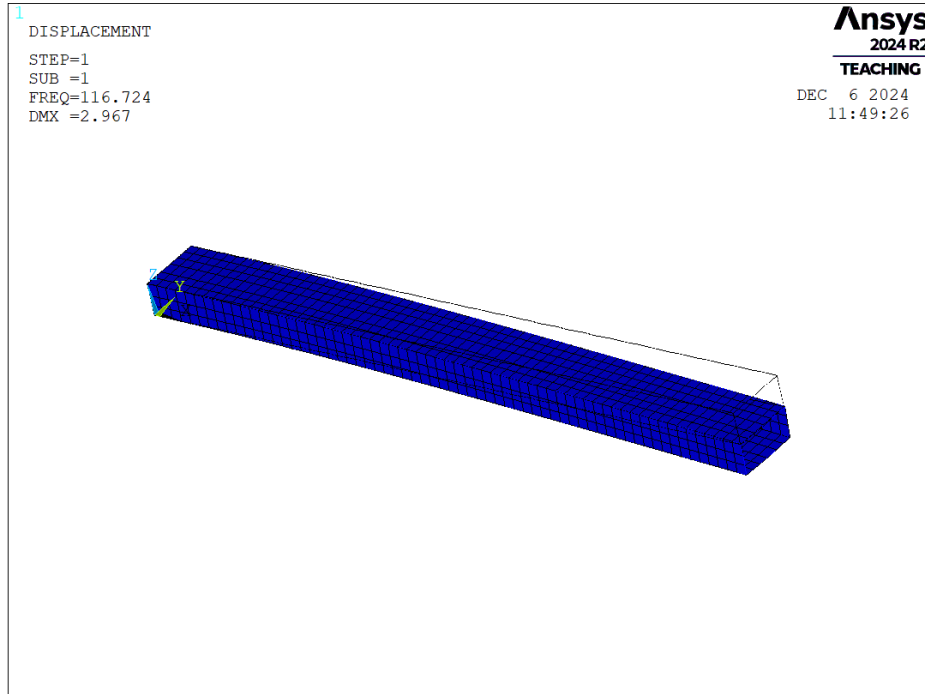


Figure 8: Part 2 - Problem 1 mode 1 shape.

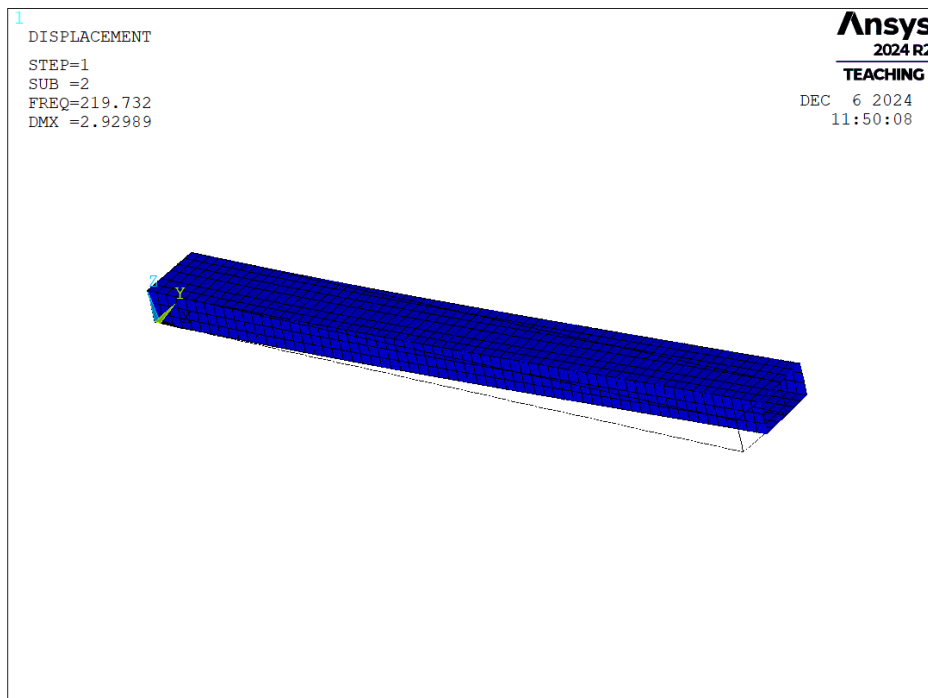


Figure 9: Part 2 - Problem 1 mode 2 shape.

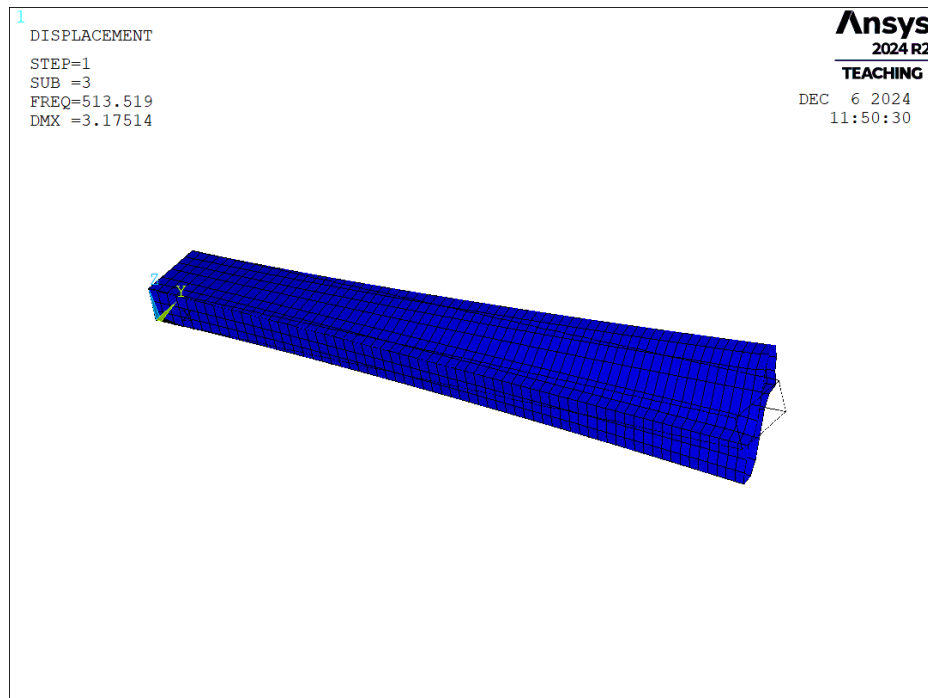


Figure 10: Part 2 - Problem 1 mode 3 shape.

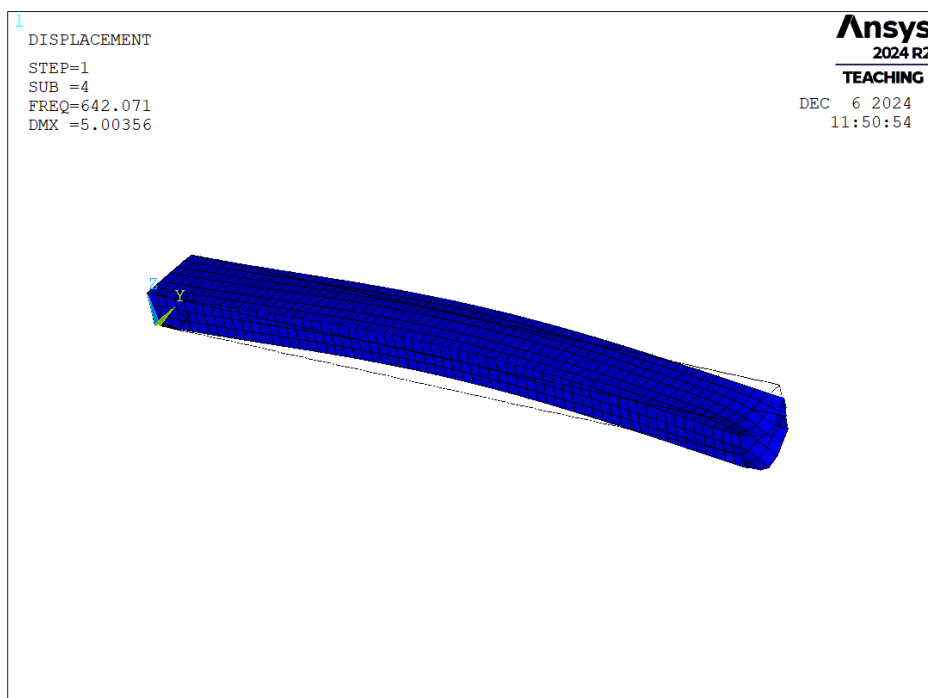


Figure 11: Part 2 - Problem 1 mode 4 shape.

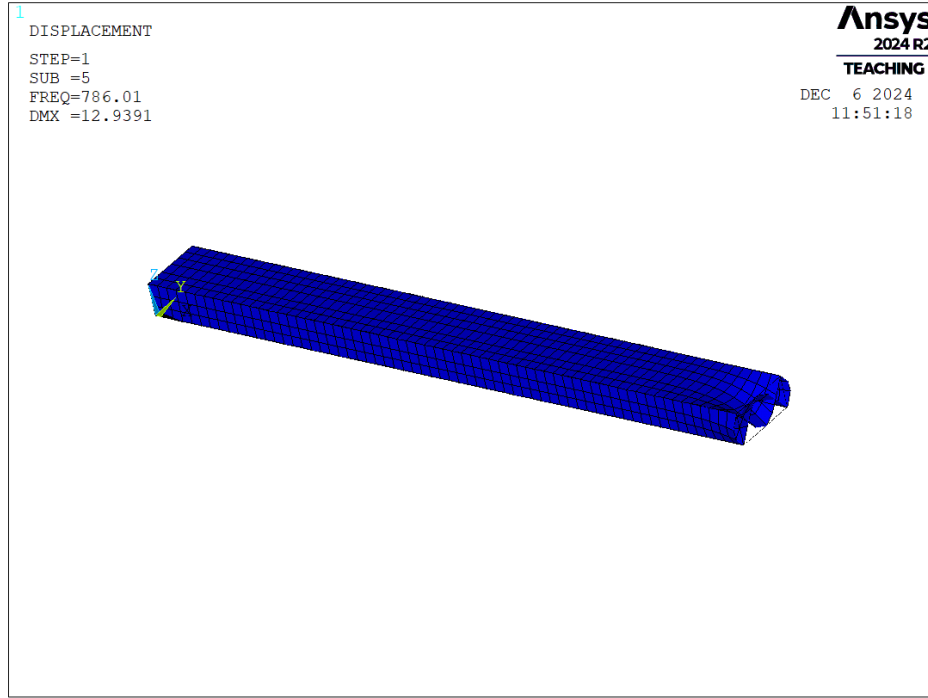


Figure 12: Part 2 - Problem 1 mode 5 shape.

Table 3: Part 2 - Problem 1 natural frequencies.

Mode	Natural frequency [Hz]
1	116.723
2	219.731
3	513.519
4	642.070
5	786.010

Problem 2: Harmonic Analysis

For the harmonic analysis, a vertical load of one is applied to the two upper unconstrained keypoints (on the right of the beam in [Figure 13](#)). The harmonic response ([Figure 14](#)) and phase angle ([Figure 15](#)) were plotted.

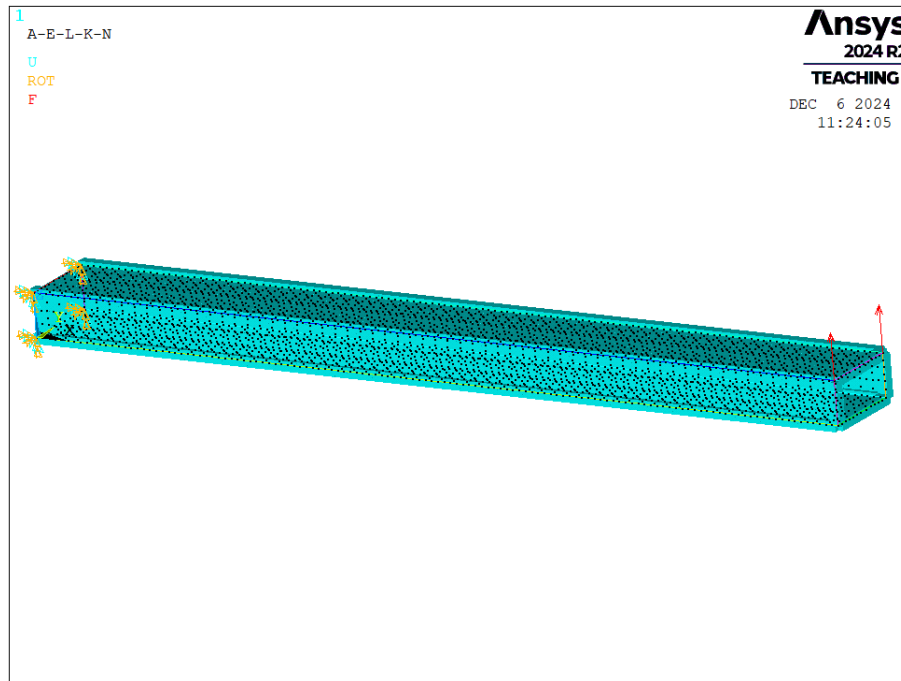


Figure 13: Part 2 - Problem 2 geometry with loads applied.

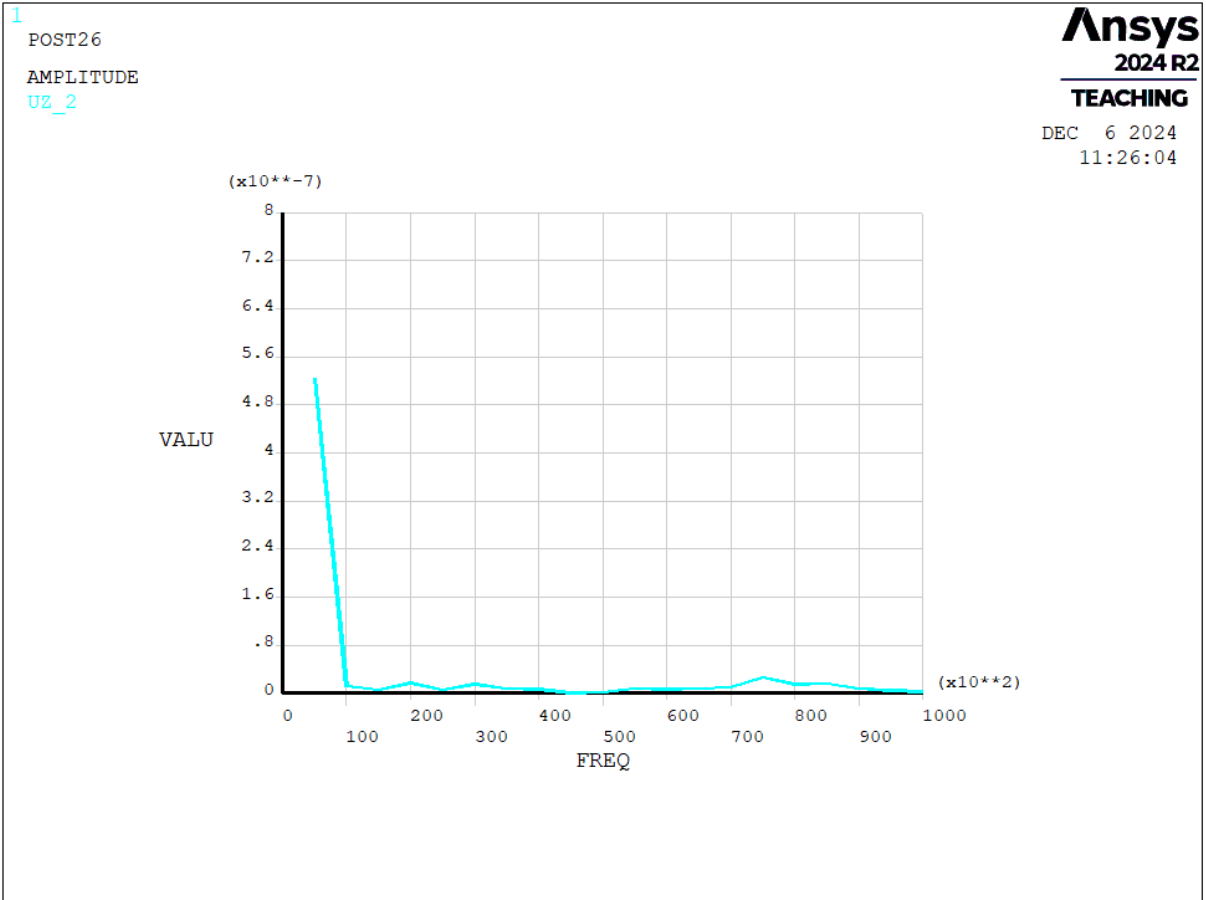


Figure 14: Part 2 - Problem 2 amplitude plot.

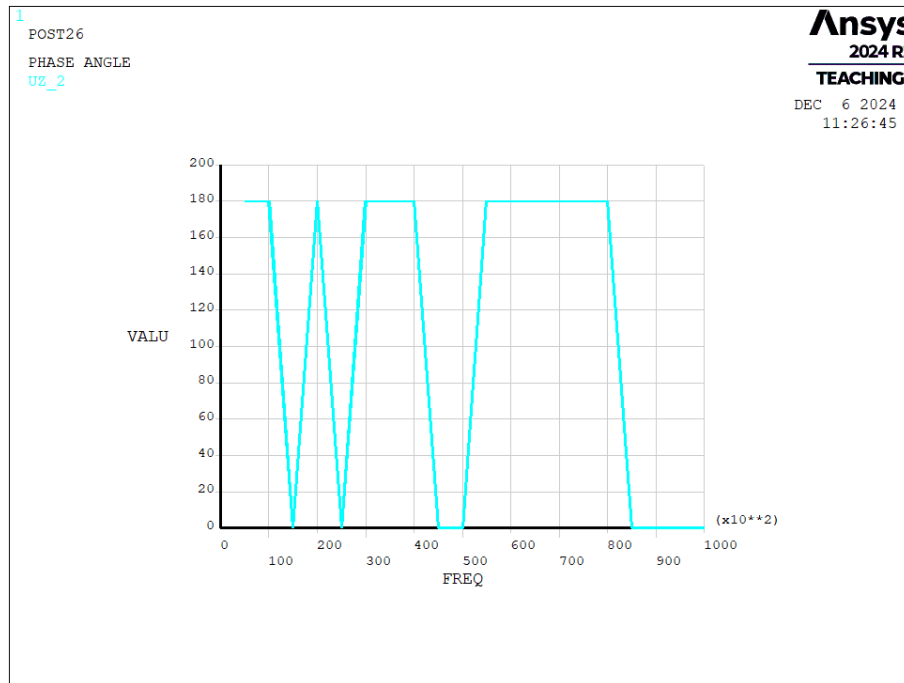


Figure 15: Part 2 - Problem 2 phase plot.

The dominant natural frequency has the greatest amplitude. From [Figure 14](#), we know that the dominant natural frequency under current loading is 5000 Hz.

In [Figure 15](#), we can understand better when the natural frequencies occur. Whenever a jump of 180 degrees (from 0° to 180° or the contrary) is observed, that represents a natural frequency.

3 Conclusion

Analyzing vibration and harmonics of a structure in Ansys allows us to learn more about the natural frequencies and mode shapes of a structure. We were first able to model and analyze a tuning fork and adjust its length to get a first resonance frequency of 440 Hz. The concept of changing a design to accommodate a certain natural frequency can apply to structures other than a tuning fork, making this a useful analysis. By plotting the amplitude and phase of the structure we can find the dominant natural frequency and its harmonics.

4 References

1. [Lab 7 Manual](#)
2. [Ansys Commands Reference Page](#)

5 Appendix A: APDL Scripts

5.1 Part 1

```
/CLEAR
/BATCH
/PREP7
/RGB,INDEX,100,100,100, 0
/RGB,INDEX, 80, 80, 80,13
/RGB,INDEX, 60, 60, 60,14
/RGB,INDEX, 0, 0, 0,15
/REPLOTT
ET,1,PLANE182,,,2
MP,EX,1,190e9
MP,PRXY,1,0.3
MP,DENS,1,7700
K,1,0,0,0
X = 0.04055
!X = 0
!
K,2,0.1,0,0
K,3,0.13,0.0275,0
K,4,0.16,0.0275,0
K,5,0.16,0.0325,0
K,6,0.13,0.0325,0
K,7,0.1,0.06,0
K,8,%X%,0.06,0
K,9,%X%,0.055,0
K,10,0.1,0.055,0
K,11,0.1,0.005,0
K,12,%X%,0.005,0
K,13,0.13,0.03,0
K,14,0.1,0.03,0
K,15,0.125,0.03,0
K,16,%X%,0,0
L,16,2 ! line 1
L,3,4 ! line 2
L,4,5 ! line 3
L,5,6 ! line 4
LARC,2,3,14,0.03
LARC,6,7,14,0.03
L,7,8 ! line 5
L,8,9 ! line 6
L,9,10 ! line 7
LARC,10,15,14,0.025
```

```

LARC,15,11,14,0.025
L,11,12 ! line 8
L,12,16 ! line 9
AL,ALL
SMRTSIZE,1
AMESH,ALL
/SOLU
ANTYPE,2
MODOPT,LANB,10
EQSLV,SPAR
MXPAND,10,,1
LUMPM,0
PSTRESS,0
MODOPT,LANB,10,20,20000,,OFF
SOLVE

```

5.2 Part 2

```

/CLEAR
/BATCH
/PREP7
/RGB,INDEX,100,100,100, 0
/RGB,INDEX, 80, 80, 80,13
/RGB,INDEX, 60, 60, 60,14
/RGB,INDEX, 0, 0, 0,15
/REPLOT
ET,1,SHELL281
KEYOPT,1,8,1
ET,2,LINK180
MP,EX,1,70e9 ! Material 1, skin material
MP,PRXY,1,0.29 ! Material 1, skin material
MP,DENS,1,2700 ! Material 1, skin material
MP,EX,2,71.4e9 ! Material 2, stiffener material
MP,PRXY,2,0.31 ! Material 2, stiffener material
MP,DENS,2,2860 ! Material 2, stiffener material
SECTYPE,1,SHELL
SECDATA,1e-3,1,0,3
SECTYPE,2,LINK
SECDATA,0.35e-4
K,1,0,0,0
K,2,0,0.05,0
K,3,0,0.05,0.025
K,4,0,0,0.025
K,5,0.5,0,0.025
K,6,0.5,0,0

```

```

K,7,0.5,0.05,0
K,8,0.5,0.05,0.025
L,1,2 ! line 1
L,2,3 ! line 2
L,3,4 ! line 3
L,4,1 ! line 4
L,5,6 ! line 5
L,6,7 ! line 6
L,7,8 ! line 7
L,8,5 ! line 8
L,1,6 ! line 9
L,4,5 ! line 10
L,3,8 ! line 11
L,2,7 ! line 12
AL,4,5,9,10 ! area 1
AL,3,11,8,10 ! area 2
AL,2,12,7,11 ! area 3
AL,1,12,6,9 ! area 4
ASEL,S,AREA,,ALL
AATT,1,,1,,1
ALLSEL
SMRTSIZE,1
AMESH,ALL
LSEL,S,LINE,,9
LSEL,A,LINE,,10
LSEL,A,LINE,,11
LSEL,A,LINE,,12
LSEL,A,LINE,,5
LSEL,A,LINE,,6
LSEL,A,LINE,,7
LSEL,A,LINE,,8
LATT,2,,2,,,2
LMESH,ALL
ALLSEL
DK,1,ALL
DK,2,ALL
DK,3,ALL
DK,4,ALL
Vibration analysis
/SOLU
ANTYPE,2
EQLV,SPAR
MXPAND,5,,,1
LUMPM,0
PSTRESS,0

```



```
MODOPT,LANB,5,20,1000000,,OFF
SOLVE
! Add load and perform harmonic analysis
! FK,5,FZ,1
! FK,8,FZ,1
! /SOLU
! ANTYPE,3
! HROPT,FULL
! HROUT,OFF
! LUMPM,0
! EQSLV, ,0,
! PSTRES,0
! OUTPR,ALL,1,
! HARFRQ,0,100000,
! NSUBST,20,
! KBC,1
! SOLVE
! FINISH
```